Spectral Representation $$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn}; \quad n = 0, 1, ..., N-1$$

Convolution $$y[n] = x[n] * h[n] = \sum_m x[m]h[n-m]$$

Cross Corelation $$y[n] = x[n] \otimes h[n] = \sum_m x[m]h[m-n]$$

Fundamental Frequency F) $\longrightarrow$ Auto-Corelation

Discrete Fourier Transform $$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}; \quad k = 0, 1, ..., N-1$$

Nyquist Rate $\longrightarrow$ 2*F0 < Fs $\quad F_{max} = F_s/2$

STFT STFT

$$X[k,m] = \sum_{n=0}^{N-1} x[n]w[n-mH]e^{-j\frac{2\pi}{N}kn}; \quad k = 0, 1, ..., N-1$$

Space : Providing Support and distinct Identity $\longrightarrow$ Machines - Detectors/Sensors

Feature Extraction $\longrightarrow$ Invariant , Distinguishing , minimal

Video : Edges , Corners , Blobs

Audio : Pitch , Loudness , Timbre

Process Followed During Machine Learning

1. Data Collection
2. Data Annotation/Labelling
3. Design Extraction
4. Model Selection
5. Testing
6. Industrial Checking

$\longrightarrow$ Data : Train,Test $\longrightarrow$ Validation Data from Train Data Used in Evaluation

Classification $\longrightarrow$ Linear , Non-Linear(Neural Network)

Soft Classification (using activation function), Hard Classification(either 0 or 1)

Quality Of Classification

Confusion Matrix

	NEGATIVE	POSITIVE
NEGATIVE	True Neg.	False Pos.
POSITIVE	False Neg.	True Pos.

Accuracy $\longrightarrow$ Correct / All

Recall $\longrightarrow$ TP / (TP+FN)

Precision $\longrightarrow$ TP / (TP + FP)

ROC curve $\longrightarrow$ Graph Between Precision and Recall

F score $\longrightarrow$ Harmonic Mean P and R

Linear Regression

1. Prediction Continuous equally likely polynomial models

$$y = w_0 + w_1 x + w_2 x^2 + ... + w_D x^D$$

$y = \Phi w \longrightarrow$ Estimationg w optimising Loss Function

Loss Function

1. Square Loss Function
2. L2 regularization (providing penaltiy (PP))
3. Lasso Regularization(PP)

$$E = (t - \Phi w)^T (t - \Phi w)$$

$$E = (t - \Phi w)^T (t - \Phi w) + \lambda w^T w$$

2

Optimizing w $\longrightarrow$

1. Analytically Solvable
2. Gradient Descent

$$w = (\Phi^T \Phi)^{-1} \Phi^T t \longrightarrow 1$$

$$w^{(\tau+1)} = w^{(\tau)} - \eta \frac{\partial E}{\partial w} \longrightarrow 2$$

Data Normalization - Zero Mean and Unit Variance

$$y^T = \sigma(\Phi(x) W_1) W_2$$ Non - Linear Regression

Sequence Of Non _ linear Regression $\longrightarrow$ Neural Network

$$y_{i_2} = \sum_{i_1=1}^{N_1} w_{i_2 i_1} \sigma\left(\sum_{i_0=1}^{N_0} w_{i_1 i_0} x_{i_0}\right)$$ Forward Pass

x_{i_0} $\quad h_{i_2} = \sum_{i_1} w_{i_2 i_1} v_{i_1}$ $\quad h_{i_3} = y_{i_3} = \sum_{i_2} w_{i_3 i_2} v_{i_2}$

$v_{i_2} = \sigma(h_{i_2})$

Loss Function = Square Loss $\longrightarrow$ $$E = \sum_n \sum_{i_2} (t_{i_2} - y_{i_2})^2$$

Optimized Using GD $$w_{ij}^{(new)} \leftarrow w_{ij}^{(old)} - \eta \left.\frac{\partial E^{(n)}}{\partial w_{ij}}\right|_{w^{(old)}}$$

Gradients Found Using Back Propogation Algorithm

$$\frac{\partial E}{\partial w_{i_1' i_1}} = v_{i_1'} \cdot \delta_{i_2'}$$

$$\delta_{i_2'} = \sigma(h_{i_2'})(1 - \sigma(h_{i_2'})) \cdot \left(\sum_{i_3} w_{i_3 i_2'} (\delta_{i_3})\right)$$

e_{i_3}

Types Of Classifiers

Multi Class Classifer

One - Hot Vector

Loss Function :
Categorical Cross Entropy

$$E_{Xent} = -\sum_c t_c \log(y_c)$$

$$y_c = \frac{e^{h_c}}{\sum_{c'} e^{h_{c'}}}$$

Multi Label Classifier

Multi - Hot Vector

Loss Function:
Binary Cross Entropy

$$-\sum_l (t_l \log y_l + (1 - t_l) \log (1 - y_l))$$

$$y_l = \text{sigmoid}(h_l)$$

Cohen's Kappa Metric for Data Annotation:

$$\kappa = \frac{P(y = y') - P(y)P(y')}{1 - P(y)P(y')}$$

$$p_y(y) = \sum_i \frac{p_x(x)}{\left|\frac{dy}{dx}\right|}\Bigg|_{x_i}$$

Tranformation Of Variables

$$\text{cov}[\boldsymbol{x}, \boldsymbol{y}] = \mathbb{E}[(\boldsymbol{x} - \mathbb{E}[\boldsymbol{x}])(\boldsymbol{y}^T - \mathbb{E}[\boldsymbol{y}^T])]$$

$$\mathbb{E}[f(x)] = \int f(x)p(x)dx$$

$$E[f(x)] \approx \frac{1}{50}\sum_i f(s_i)$$

Samples vs Pdf

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Univariate Gausian Distribution

$$(2\pi)^{-\frac{k}{2}} \det(\boldsymbol{\Sigma})^{-\frac{1}{2}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})},$$

exists only when $\boldsymbol{\Sigma}$ is positive-definite

Multi Variate Gausian distribuitoin

Clustering :

1. K_means
2. Latent Variable Models

$$\mathcal{L} = \sum_{n,k} r_{nk} \|x_k - \mu_k\|^2$$

Optimal Value of K-means Cluste

$$\mu_{k'} = \frac{\sum_n r_{nk'} x_{k'}}{\sum_n r_{nk'}}$$

:

$$r_{nk} = \begin{cases} 1, & \text{if } k = \underset{k'}{\text{argmin}} \|x_n - \mu_{k'}\| \\ 0, & \text{o.w.} \end{cases}$$

2 $$p(x) = \sum_z p(x|z)p(z)$$

Gausian Mixture Model

$$p(x) = \sum_k \pi_k \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

Parameter's Optimal

$$\mu_{k'} = \frac{\sum_n \gamma_{nk'} s_n}{N_{k'}}$$

$$\pi_{k'} = \frac{N_k}{N}$$

$$\Sigma_{k'} = \frac{1}{N_{k'}} \sum_n \gamma_{nk'} (s_n - \mu_{k'})(s_n - \mu_{k'})^T$$

Optimization

GD -> Learning Rate quick -< small and consistent , slow -<big and inconsistent

Stopping Criteria : Validation Error Stops Decreasing.
Stochastic Gradient Descent: Start with Small batch size increase with epoch

Faster Technology : Early Stopping, Drop outs

Probability :

$$P(w) = \sum_t P(t, w)$$

Marginlization

$$P(t|w) = \frac{P(w|t)P(t)}{P(w)}$$

Baye's Rule

$$P(w|t) = \frac{P(w,t)}{P(t)}$$

Conditional Rule

Parameter Estimation

1. Maximum LikeliHood Estimation
2. Maximum A-priori Estimation

$$\hat{\theta}_{MAP} \text{ argmax}_\theta P(x|\theta)P(\theta)$$

$$\hat{\theta}_{ML} = \text{argmax}_\theta P(x|\theta)$$

For Gauss

$$\hat{\sigma}^2_{ML} = \frac{1}{N}\sum_{i=1}^{N}(s_i - \hat{\mu}_{ML})^2$$

$$\hat{\mu}_{ML} = \frac{1}{N}\sum_{i=1}^{N} s_i$$

Sufficient Statistics :Minimum Amount of statistics required for variables

Sample From A given Pdf

$$y = CDF^{-1}(x)$$

EM Algo ;Followed in Clustering

Distance IN pdf's:
Cross Entropy
KL divergence

$$-\sum_x p(x) \log(q(x))$$

$$-\sum_x p(x) \log \frac{q(x)}{p(x)}$$

KL divergence is used in all places wheras Cross entropy in in places where one pdf is delta function

Mutual Information : I(X:y = H(X) - H(X/Y)

EM algo

$$\log p(X|\theta) \geq \sum_Z q(Z) \log \frac{p(X, Z|\theta)}{q(Z)} = \mathcal{L}(q(Z), \theta)$$

$$\log p(X|\theta) - \mathcal{L}(q(Z), \theta) = \text{KL}\,[q(Z)||p(Z|X, \theta)]$$

$$q \leftarrow \text{argmin}_q \text{ KL}\,[q(Z)||p(Z|X, \theta)]$$ E - Step

$$\theta \leftarrow \text{argmax}_\theta \mathbb{E}_q[\log p(X, Z|\theta)]$$ M-step

PCA - Dimensionality reduction and Matrix Factorization

Normalize: $\boldsymbol{x}_n = \boldsymbol{s}_n - \mathbb{E}[\boldsymbol{s}]; \quad \mathbb{E}[\boldsymbol{s}] = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{s}_n$

Obtain variance matrix $S = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n \boldsymbol{x}_n^T$

Eigen value decomposition of S to get $\lambda_i, \boldsymbol{u}_i; i = 1, \ldots$ decreasing order

Choose first M eigen vectors as the principal axes

$$\tilde{\boldsymbol{x}}_n = \sum_{i=1}^{M} (\boldsymbol{x}_n^T \boldsymbol{u}_i) \boldsymbol{u}_i$$

$$Y = U^T X$$ Transformed Vector

PLCA → Classification

$$q_t(z|f) = \frac{P_t(z)\, P(f|z)}{\sum_z P_t(z)\, P(f|z)}$$ E - STEP

M - Step

$$P_t(z) = \frac{\sum_f V_{ft}\, q_t(z|f)}{\sum_{z'}\sum_f V_{ft}\, q_t(z'|f)}$$

$$P(f|z) = \frac{\sum_t V_{ft}\, q_t(z|f)}{\sum_f \sum_t V_{ft}\, q_t(z|f)}$$

Non Negative Factorization

Loss Function

$$F(h) = \sum (V_i - Wh_i)^2$$

$$K_{ab}(h^t) = \delta_{ab}(W^T W h^t)_a / h_a^t$$

$$G(h, h^t) = F(h^t) + (h - h^t)^T \nabla F(h^t) + \frac{1}{2}(h - h^t)^T K(h^t)(h - h^t)$$

$$W_{ir} \leftarrow W_{ir} \frac{(VH^T)_{ir}}{(WHH^T)_{ir}}$$

$$H_{rj} \leftarrow H_{rj} \frac{(W^T V)_{rj}}{(W^T W H)_{rj}}$$

Updation Rule